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A hybrid bio-inspired model for predicting urban air pollution using deep learning

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Accurate prediction of urban air quality is vital for protecting public health and the environment. The current challenges in air quality prediction models include noisy data, missing data, interactions of pollutants, temporal and spatial variations, external variables, lack of generalization, and real-time prediction. To overcome these challenges, a hybrid bio-inspired model for predicting urban air pollution using deep learning (AQP-SAPINN-HMRFO) is proposed. The input data is obtained from the Global Urban Air Quality Index dataset. The data is preprocessed with Implicit Bulk Surface Filtering (IBSF) to normalize the data and treat missing values to provide high-quality input data. The Quadratic-Phase Wave Packet Transform (QPWPT) is used to extract relevant features such as concentrations of pollutants, interactions between pollutants, and past concentrations of pollutants. Air quality forecasting is carried out by using a Self-Adaptive Physics-Informed Neural Network (SAPINN) that predicts concentrations of five major air pollutants like Particulate Matter $\mathrm{PM}_{2.5}$, Carbon Monoxide (CO), Nitrogen Dioxide (NO_2), Ozone (O_3), and Sulfur Dioxide (SO_2), along with meteorological factors like temperature, wind speed, and humidity at various locations. The Hierarchical Manta Ray Foraging Optimization (HMRFO) technique is used to optimize the weight parameters of SAPINN to improve the accuracy of the model. The AQP-SAPINN-HMRFO model is a combination of SAPINN and HMRFO techniques. This model is capable of handling the interactions and gaps of the pollutants effectively. The proposed method is implemented in Python and the experiments demonstrate that AQP-SAPINN-HMRFO achieves 99% accuracy. This is a significant improvement over existing approaches and indicates its potential for real-time applications in urban air quality monitoring, environmental management, and strategic urban planning.

Keywords Hierarchical Manta Ray Foraging Optimization Algorithm, Implicit Bulk-Surface Filtering, Quadratic-Phase Wave Packet Transform, Self-Adaptive Physics-Informed Neural Networks, Urban Air Quality Index, Environmental Monitoring.

The rapid expansion in population worldwide and air pollution issues have led to strain over the environment and human being health in recent years[1]. There are several harmful effects on human health due to increased concentrations of gaseous compounds, such as $\mathrm{PM}_{2.5}$, CO, NO_2, O_3, and SO_2 in the environment[2–4]. Industrial emissions, transportation pollution, and climatic components such as precipitation, wind speed, temperature, and relative humidity all have a significant impact on urban air quality[5,6]. Despite its substantial impact on human health, the prediction of gaseous pollutants has gotten relatively less attention than particulate pollution forecasting, which is typically the focus of most study[7–10]. The development of cities and spatial planning is important in determining the air quality conditions in urban areas[11]. Development in rapid city is likely to amplify transportation, industrial, and energy consumption emission, consequently affecting both temporal and spatial distribution of atmospheric pollutants[12]. Moreover, the environmental conditions (e.g. the geography and distribution of green spaces) may alter the dispersion and accumulation of pollutants[13]. These interactions are important to consider comprehending and controlling the process of air quality in cities[14]. $\mathrm{PM}_{2.5}$ is associated with respiratory diseases such as chronic obstructive pulmonary disease (COPD), asthma, and bronchitis, as

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well as cardiovascular diseases including heart attacks and stroke. O_3, NO_2, and SO_2 increase the risk of COPD hospitalizations, lung inflammation, and reduced lung function. Long-term exposure to these pollutants has also been linked to premature mortality, oxidative stress, and systemic inflammation, highlighting the critical need for accurate air quality prediction and management to protect public health[15,16]. Accurate air quality prediction is necessary for the betterment of the environment as well as human health and is essential for government decision making. However, it is challenging to forecast air quality using the concentration of harmful gases[17]. Deep learning models are employed more in the big data and artificial intelligence domains lately[18–20] . The deep learning ability to extract intrinsic features from a vast amount of input data and develop efficient feature representations makes it a popular choice among researchers[21,22]. The gaps in the previous research include:

- Modern time series techniques including the combined spatio- temporal model with auto-regression, only account for one of the temporal and spatial relationships, overlooking the intricate interplay between metrological data at various geographic locations and pollutant concentrations over time. Additionally, incorporating data from unrelated monitoring sites can lower prediction accuracy.
- Existing deep learning models models primarily employs basic pre processing techniques and not robust to noise and outliers. The current methodlogoies often disregard advanced techniques that adapt better to sparsity, and reconstructing underlying patterns more accurately.
- Traditional forecasting models face a significant limitation in their ability to generalize across diverse urban profiles. Various factors such as diversity in city infrastructures, traffic management, environmental conditions, and pollution sources present substantial challenges, hindering the accuracy and applicability of these models across different urban contexts
- Current methodologies struggle to effectively manage data imbalances in air quality datasets. The uneven distribution of pollutant concentrations and associated meteorological variables across geographic regions calls for hybrid approaches to ensure accurate and unbiased forecasting results.

Therefore, the researchers have put forth a number of innovative hybrid models for air quality prediction to further increase the model's prediction accuracy. The motivation behind this work is to develop a more appropriate and reliable air quality prediction method that captures the complex spatio-temporal relationships among multiple pollutants and environmental factors.

The innovation of the proposed AQP-SAPINN-HMRFO method lies in the synergistic combination of advanced preprocessing, feature extraction, physics-aware neural network modelling and hierarchical optimization for accurate air quality prediction.

The main contribution of this research work is summarized below:

- The innovation of the proposed AQP-SAPINN-HMRFO method lies in the synergistic combination of advanced preprocessing, feature extraction, physics-aware neural network modelling and hierarchical optimization for accurate air quality prediction.
- Self adaptive physics informed neural network(SAPINN) introduces domain knowledge into the learning process by embedding physical constraints, while its adaptive mechanism allows dynamic tuning based on data variability for improved generalization.
- To further enhance predictive accuracy, the hierarchical manta ray foraging optimization algorithm(HMRFO) is adopted to optimize SAPINN's weight parameters, offering superior global search capabilities through multi-level foraging behaviors, increasing overall accuracy and robustness.

The manuscript is arranged as follows: "Literature review" section discusses the prior work related to air quality prediction with comparison, the section "Proposed methodology" explains the proposed technique and methodology including pre-processing, feature extraction and neural network, followed by "Results" section that provides the experimental results with performance analysis, ablation study, and statistical analysis. Paper is concluded with section "Conclusion" and discusses the conclusion and future work.

Literature review

Several investigation works were presented in literature based on air quality prediction using neural networks and related techniques. This section discusses and compares various methods for air quality prediction using deep learning and other techniques.

In 2021 Seng, D.et al[23] introduced LSTM neural network-based spatiotemporal air quality prediction model using the time series information on air quality gathered from 35 Beijing air quality monitoring sites. To calculate the air quality indicator concentration values at the usual stations, the model was fed the air quality sequences from the stations with dissimilar information features. The total AQP (air quality prediction) result for Beijing was calculated using the calculated average values. The method performance was verified against two of the most sophisticated models as well as other baseline models. The experimental outcomes depict that the suggested model outperforms alternative baseline models. It achieves greater accuracy and lesser sensitivity.

In 2021 Zhang, L et al.[24] demonstrated use of semi-supervised bidirectional LSTM-Neural network for air quality prediction. Considered as signal data,$PM_{2.5}$ time-series data were the sole inputs required. EMD was an unsupervised feature learning technique used to break down data and eliminate frequency characteristics, such as amplitude. The method enhanced the forecast of short-term trends, particularly for abrupt modifications. During the managed learning phase, BiLSTM was employed. Five datasets gathered from China National Environmental Monitoring Centre were utilized to verify a suggested model's predictive ability. It attains high sensitivity and low accuracy.

In 2021 Espinosa, R., et al.[25] introduced a multicriteria approach to air quality prediction based on time series forecasting. CNN, GRU, and LSTM were paired using various window sizes, and the Forest, Lasso, and Provision Vector Machine regression examples were inspected. Therefore, the best models provide a very accurate 24-hour forecast of the air pollution concentration in the area under consideration. It attains a high f1-score and low specificity.

In 2022 Xiao, C., et al.[26] presented dual-path dynamic directed graph convolutional network for predicting air quality. The study enables a comparison between the dynamic associations created by wind speed and direction and the stations' static distance correlations. A dual-path dynamic directed graph was used to more thoroughly depict the dynamic spatial dependencies. Gated recurrent units (GRUs) extract complex temporal correlations of historical air quality data and integrate future weather patterns. The dynamic spatiotemporal gated recurrent block was constructed utilizing dual-path dynamic directed graph blocks. The data with $\mathrm{PM}_{2.5}$ concentration from real-world datasets were used. It attains high throughput and low mean absolute error.

In 2023, Ravindiran, G., et al.[27] introduced air quality prediction by machine learning models: a predictive study on the indian coastal city of Visakhapatnam. When compared to the traditional methods, machine learning has become a viable methodology for estimating the Air Quality Index (AQI). It applies the AQI to the city of Visakhapatnam, Andhra Pradesh, India, focusing on 12 contaminants and 10 meteorological parameters from July 2017 to September 2022. For this purpose, it applied many machine learning models, including LightGBM, Random Forest, Catboost, Adaboost, and XGBoost. It attains high f1-score and low sensitivity.

In 2025, Rajesh, M., et al.[28] presented machine learning driven method for real time air quality assessment and predictive environmental health risk mapping. The suggested paper draws on a number of data sources, like fixed air quality sensors, mobile air quality sensors, additional meteorological inputs, satellite data, localised demographic information. The proposed system uses machine learning strategies - Random Forest, Gradient Boosting, XGBoost, and Long Short Term Memory (LSTM) networks - to identify concentrations of pollutant and classify air quality into categories with very good temporal resolution in their predictive ability. It interprets predictions using SHAP analysis to uncover the important environmental and demographic variables that contributed to each prediction. It attains low root mean squared error and high f1-score.

In 2024, O'Regan, A.C., et al.[29] introduced the data fusion for improving urban air quality modeling utilizing large-scale citizen science data. By combining a dispersion model output with extensive citizen science data gathered over a 4-week period by 642 people in Cork City, Ireland, it improves urban air quality modeling. The dispersion model filled in the gaps in regulatory monitoring efforts and made it possible to identify the main sources of NO_2 air pollution. It created a data fusion model that captured localized variations in air quality by combining the diffusion tube data with the output of the dispersion model. A more accurate evaluation of the population exposure to air pollution was made possible by this improved accuracy. It attains greater specificity and lesser accuracy. Table 1 shows the comparison of the literature survey,

There are studies that have helped us predict the air quality using neural networks and machine learning. However air quality prediction still has some problems. LSTM-based models by Seng et al. (2021)[23] utilized LSTM for spatiotemporal prediction in Beijing but depended on averaged values from stations, thus limiting generalization. Zhang et al. (2021)[24] used BiLSTM with EMD to forecast short term $\mathrm{PM}_{2.5}$ but ignored multi, pollutant interactions. Espinosa et al. (2021)[25] used CNN, GRU, and LSTM together but incorporated only inadequately meteorological variables. Xiao et al. (2022)[26] used graph convolutional networks but did not perform parameter optimization. Ravindiran et al.(2023)[27] and Rajesh et al. (2025)[28] predicted AQI using machine learning but without adaptive physics, informed modeling. O'Regan et al.(2024)[29] enhanced local exposure through data fusion but did not focus on prediction accuracy. The AQP-SAPINN-HMRFO method eliminates the drawbacks of earlier research work by first integrating IBSF which is a highly effective data preprocessing technique capable of handling missing values, noise, and outliers to provide clean and high, quality inputs. QPWPT not only extracts comprehensive multi, pollutant features but also captures through single time series pollutant concentrations, their interactions and changes over the past, which are the three main aspects of air

Author(s)	Objectives	Methods	Merits	Demerits
Seng, D. et al.[23]	Analyze spatiotemporal features for air quality across 35 Beijing stations	LSTM neural network using dissimilar station data	The model effectively captures spatial diversity by utilizing data from multiple monitoring stations, improving overall prediction accuracy	Using dissimilar feature information from unrelated stations may introduce noise and reduce localized prediction precision
Zhang, L. et al.[24]	Short-term $\mathrm{PM}_{2.5}$ trend prediction	Semi-supervised BiLSTM + Empirical Mode Decomposition (EMD)	Enhances short-term PM2.5 trend prediction during sudden changes	Ignores influence of other pollutants and environmental variables
Espinosa, R. et al.[25]	Comparative analysis of pollution forecasting models	CNN, GRU, LSTM + RF, Lasso, SVM with multicriteria evaluation	Supports informed decision-making to reduce population health impacts	Requires extensive computational resources and tuning
Xiao, C. et al.[26]	Model dynamic spatiotemporal air quality dependencies	Dual-Path Dynamic Directed GCN + GRU	Captures dynamic spatial dependencies using wind-based dual-path graphs	Increased model complexity and computational load
Ravindiran, G. et al.[27]	Predict air quality in Visakhapatnam	Machine learning: LightGBM, Random Forest, CatBoost, Adaboost, XGBoost	Handles multiple contaminants and meteorological parameters; high F1-score	Limited interpretability; focused on one city only
Rajesh, M. et al.[28]	Real-time air quality estimation and health risk mapping	Random Forest, Gradient Boosting, XGBoost, LSTM; SHAP for interpretability	Integrates multiple data sources; provides interpretability and actionable health risk maps; high temporal accuracy	Complex architecture; requires cloud infrastructure and continuous data streams
O'Regan, A.C. et al.[29]	Enhance urban air quality modeling utilizing citizen science	Data fusion of dispersion model output + citizen science measurements	Captures localized fluctuations; higher specificity; informs public health policies	Slightly lower overall accuracy; limited duration (4-week study

Table 1. Summary of literature survey.

pollution, that are key to the correct forecasting of future pollution levels. SAPINN takes advantage of physics, informed neural networks (PINNs) for incorporating environmental laws into the model thereby guaranteeing predictions that are both physically consistent and realistic. The HMRFO method finds the optimal solution of the weight parameters in SAPINN so as to improve the speed of convergence, stability and the capacity for generalization to be able to handle a variety of urban conditions effectively. In summary, this method represents a powerful and efficient way to tackle air pollution prediction issues.

Proposed methodology

This section describes the proposed AQP-SAPINN-HMRFO method. Initially, data is preprocessed utilizing an implicit bulk surface filter (IBSF), followed by feature extraction using quadratic phase wave packet transform (QPWPT), and finally, the prediction step uses a self-adaptive physics informed neural network(SAPINN) for predicting the air quality. The final stage uses the HMRFO to optimize and fine-tune the neural network parameters for improved performance. This thorough approach guarantees better data quality, focused analysis, and the best possible use of extracted features, resulting in a reliable and effectual system for the prediction of air quality. Fig. 1 represents a block diagram of the proposed AQP-SAPINN-HMRFO technique.

The comprehensive explanation about each step is provided.

Data acquisition

The input data is collected from global urban air quality index dataset[30]. This dataset presents a complete air quality index (AQI) data from major global cities over the years 2015 to 2025, gathered from monitoring stations in local governments, environmental agencies, and open APIs. Also included in the dataset are daily time-stamped AQIs along with the major pollutants $PM_{2.5}$, PM_{10}, CO, NO_2, O_3, and SO_2 and meteorological factors (temperature, humidity, and wind speed) for in-depth analysis. The dataset comprises 36, 60 valid entries and did not have any mismatched or missing values. The "Date" field shows a time stamp from 1 January 2024 to 31 December 2024 with a mean date of 1 July 2024. The City contains New York, Los Angeles, London, Beijing, Delhi, Tokyo, Paris, Sydney, São Paulo, and Cairo with New York as the most named city by 10%. The "Country" field contains USA, UK, China, India, Japan, France, Australia, Brazil, and Egypt with USA being the most mentioned country at 20%. The AQI field ranged from 30 to 300, mean AQI 165, standard deviation 78.6, and quartiles of 96, 165, and 233 respectively. The dataset resulted in 3,660 entries AQI. The data gathering consisted of data gathered from New York, London Beijing, and Delhi creating a total of 1,464 entries for training. The data gathering for LA, Tokyo, Paris, Sydney, São Paulo, and Cairo, created a total of 2,196 entries for testing and the split was 40% for training and 60% for testing.

Pre-processing using implicit bulk-surface filtering

The pre-processing using IBSF[31] along with data normalization and handling missing values is discussed in this section. It offers smoother pollutant fields, preserves meaningful boundary gradients, and accurately reconstructs missing or noisy sensor values. Combined with normalization, it enhances data consistency, improves model stability, and provides a physically reliable input for subsequent learning and prediction stages. IBSF is grounded in the physical behavior of pollutants, which disperse smoothly in open areas but change abruptly near sources and boundaries. By separating bulk and surface effects, it mimics real atmospheric interactions, enabling more

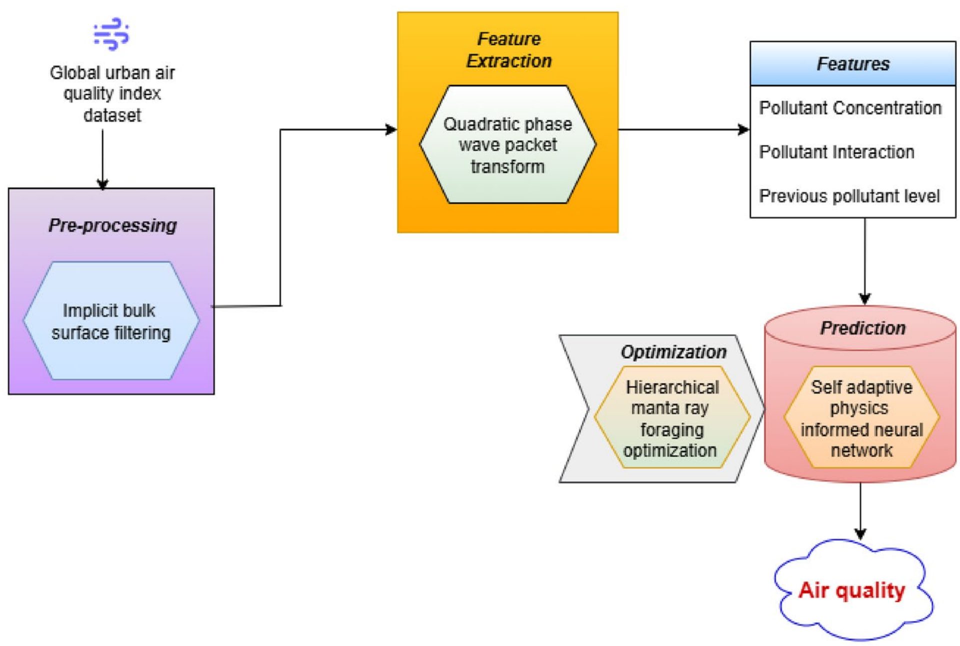


Fig. 1. Block diagram of AQP-SAPINN-HMRFO.

realistic and consistent air-quality preprocessing. IBSF is suitable for air-quality preprocessing because pollutant fields inherently display two behaviors: smooth spatial dispersion in open regions and sharp concentration gradients near emission sources, obstacles, and boundary surfaces. This closely aligns with the bulk–surface decomposition used in IBSF. Its elliptic formulation enables stable reconstruction of missing or sparse sensor values, reduces satellite-derived noise, and suppresses outliers. These properties make IBSF scientifically appropriate for pollutant and meteorological time-series preprocessing. IBSF separates urban pollutant variation into two components smooth changes in open areas and sharper gradients near boundaries such as roads and surface. A coupled filtering process is applied to capture both behaviours simultaneously by using equation (1).

$$a = (k_y^x)^1 \Delta \cdot \beta + y \tag{1}$$

where, a denotes the final bulk-filtered output, (k_y^x) denotes the bulk filtering coefficient, and Δ represents the laplacian operator, β represents the surface correction term, and y represents the data-fidelity component. Observed pollutant values are incorporated as data-fidelity terms, giving higher weight to reliable sensor readings while automatically estimating missing or corrupted values by using equation (2).

$$\beta = \mu hx(q(y)) + 2\mu q(y) \tag{2}$$

where μ implies the weighting factor, $hx(q(y))$ represents the surface operator acting on sensor-quality–based function $q(y)$.This operator reconstructs missing values by propagating information from nearby valid observations while simultaneously suppressing high-frequency noise, especially in satellite-derived pollutant attributes. The elliptic formulation ensures physically smooth reconstruction across the spatial domain. A surface component adjusts the bulk filter to preserve realistic pollutant gradients around critical boundary regions, improving accuracy where emissions or obstacles cause abrupt concentration changes by using equation (3).

$$h(k) = \frac{1}{2}\left(\Delta x + (\Delta x)^t\right) \tag{3}$$

where, $h(k)$ represents the symmetric smoothing kernel, Δx represents the spatial difference operator capturing directional variation in data and $(\Delta x)^t$ represents the transpose of the spatial difference operator. The bulk and surface components are combined and normalized to produce a smooth, physically consistent pollutant field that replaces noisy measurements and fills missing data by using equation (4).

$$U_f(x) = (I - K_b(x)\Delta)^{-1}\left[\Lambda(x)F(x)\right] + \varepsilon\left\{(I - K_b(s)\Delta_\Lambda)^{-1}\left[M(s)Q(s)\right]\right\}(x) + D(x) \tag{4}$$

where, $U_f(x)$ denote the preprocessed field, $F(x)$ represents raw observed pollutant field,$\Lambda(x)$ denotes the data-fidelity weight, $K_b(s)$ signifies the helmholtz regularizer, Δ indicates the bulk smoothing operator, $M(s)$ denotes the surface weighting on boundary, $Q(s)$ denotes the boundary-derived surface proxy, $K_s(s)$ denotes the surface lengthscale for Laplace–Beltrami regularization, Δ_Λ represent the Laplace–Beltrami operator on Λ, ε represent the extension operator that maps a surface solution on Λ, $D(x)$ represent the normalization correction. Finally by using IBSF handled the missing values of data and normalized the data. After that, the pre-processed data are fed to the feature extraction stage.

Feature extraction using quadratic phase wave packet transform

This section describes the feature extraction using QPWPT[32]. The QPWPT produced a 128-dimensional feature vector for each input window. Specifically, 48 coefficients corresponded to pollutant concentration (low- and mid-frequency components), 32 coefficients captured pollutant interaction terms derived from bi-directional time–frequency modulation, and 48 coefficients represented previous pollutant levels through high-frequency QPWPT components. Each group contributed uniquely to the prediction task such as concentration-related coefficients encoded long-term pollutant behavior, interaction coefficients modelled nonlinear dependencies among pollutants, and high-frequency coefficients captured sudden variations in historical levels. An internal validation study confirmed that removing interaction features reduced prediction accuracy, demonstrating their essential role in representing cross-pollutant relationships. Although full multi-level quaternion wavelet packet decomposition enhances feature richness, several lightweight configurations can substantially reduce computational demand with minimal impact on accuracy. In particular, the decomposition depth can be reduced (for example, from four levels to two levels), and selective node pruning can be applied to retain only the most informative sub-bands. Additionally, fixed filter banks may replace adaptive ones to lower processing overhead. These modifications allow QPWPT to operate efficiently on resource-constrained edge devices while preserving essential frequency–temporal characteristics needed for robust air-quality prediction. The QPWT's correctly capture transitory phenomena, which make it useful for applications that need high-resolution time-frequency analysis. Moreover, it provides an adaptable framework for signal processing like feature extraction, compression and denoising. Its localization characteristics reduce aliasing and spectral leakage while making it easier to retrieve valuable information from complex values. Substitute the QPWT kernel for the wave packet transform kernel and the quadratic-phase wavelet for the wavelet is expressed in equation (5),

$$v_y^x(\mu, \omega, \Delta) = \int_y f(t)\psi_{\omega,\mu}(t)x_x(y,t)\,xt \tag{5}$$

where, v_y^x represents the adjusting of the parameter,μ implies the initial random variable, ω denotes the particular time-frequency transform in the function, Δ represents the few new time-frequency instruments, $\int_y$ represents the innovative canonical linear wave packet transform, $(t)\psi_{(\omega,\mu)}$ denotes the number of initial random variables of the parameter and $(y,t)\,xt$ represents the high parameter as given in equation (6),

$$p_y^x(\mu,\omega,\Delta)=\int_y f(t)v_v=\left(\frac{h-\alpha}{\beta}\right)-e^x\frac{b}{2b}(\alpha^2-\mu^2)fx \tag{6}$$

where p_y^x denotes the pollutant wave packet transform, and (μ,ω,Δ) denotes the bi-directionality of the pollutant wave packet transform, and $h-\alpha$ represents the higher quality of the quadratic pollutant wave packet transform, and β implies the initial random variable of the pollutant wavelet, $\alpha^2-\mu^2$ represents the fundamental concentration of pollutant, as given in equation (7),

$$v_x\omega_{\Delta,\beta}(\mu)=\int_x v(x,y)f_{ex}(v)\,xy \tag{7}$$

where, $v_x\omega_{(\Delta,\beta)}(\mu)$ represents the concern regarding the internal product, ω is the initial random variable, $(y)\,xy$ signifies the pollutant of the pixel value, and (*x*, *y*) denotes the pixel value of the quadratic wave packet transform. The term $[\omega_{(\Delta,\beta)}]$ represents multiple values. The pollutant interaction is calculated as given in equation (8).

$$v_y^x(\mu,\omega,\Delta)=\left(q_x\left\lfloor\int_{\mu,\omega}\right\rfloor qx\right) \tag{8}$$

where, v_y^x implies the initial variable of the initial random variable, q_x signifies the pixel value in signal processing, $\int_{(\mu,\omega)}$ represents the signal representation of pixel values, and $(q_x\lfloor\int_{\mu,\omega}\rfloor qx)$ denotes the variable. The quadratic wave packet transform of previous pollutant levels is given in equation (9).

$$v_y^x(\mu,\omega,\Delta)=\left(y_x\left\lfloor\int_{n,m}px\right\rfloor\right) \tag{9}$$

where, v_y^x denotes the initial variable of pixel value, *px* implies the parameter of the quadratic variable, $\left\lfloor\int_{n,m}\right\rfloor$ represents the multiple representations of multiple values, and $(y_x\lfloor\int_{n,m}\rfloor px)$ denotes the variable of the quadratic wave packet transform. Finally, the features, such as pollutant concentration, pollutant interaction, and previous pollutant levels, are extracted. Then the extracted outputs are fed to the air quality prediction process.

Air quality prediction using self-adaptive physics-informed neural networks

In this section, Air Quality Prediction using SAPINN[33] is discussed. The objective of SAPINN for predicting air quality as concentration of five gaseous substances namely $\mathrm{PM}_{2.5}$, CO, NO_2, O_3 and SO_2.The network integrates the fundamental physical laws governing atmospheric processes, such as pollutant dispersion and chemical reactions, directly into their architecture, ensuring predictions remain scientifically plausible and improving overall accuracy. A key benefit is self-adaptive capability, which dynamically optimizes model parameters during training to maintain an optimal balance between data fidelity and physical consistency, particularly valuable for handling the dynamic nature of air quality systems. This combination ensures that solutions follow physical laws, requires less data, and produces the accurate forecasts. The SAPINN resilience to ambiguity and adaptability during training allows them to scale and generalize well to real-world situations. The primary goals are to establish self-adaptive PINNs as a viable method for tackling difficult physics-based issues in a range given in equation (10),

$$A(\Delta,a_1,a_2,a_0)=A_5(\Delta)+\Big(A_x(a,A_1)+\big(A_x(a,A_1)+A_0(a,A_0)\big)\Big) \tag{10}$$

Here, A signifies air, Δ represents the capable of training, represents the constructive self-adaptation, $(\Delta,a_1\cdot a_2,a_0)$ represents the quality of air, $A_x(a,A_1)$ represents the weights related to the initial, $A_s(\Delta)$ represents the initial random variable of self-adaptation, in which the network weight and $(A_x(a,A_1))$ represents the loss function's side-by-side representation. The reliability of succeeding layers in providing scientifically reasonable air quality projections. For instance, if a sensor erroneously reports an elevated $\mathrm{PM}_{2.5}$ value due to calibration drift, the input layer detects and fixes such abnormalities before they propagate through the network. The gradient descent is given in equation (11),

$$q(a,h_x)=\frac{1}{2}\sum_{j=1}^{mn}q_x(h_x)M_{y,x}\left[\mu(y_j,x_j';\mu)\right]-h(y_j,x_j')^2 \tag{11}$$

where q represents the quality of air, and a, h_x represents the fundamental functions of self-adaptation, $\sum_{j=1}^{mn} n(h_x)$ represents the reduced values compared to those of the network, $M_{y,x}$ denotes the non-detrimental function, and (y_j^i, x_j^i) represents the initial random variables at the borders and residue spots. These hidden layers perform advanced feature extraction by identifying patterns such as CO levels increasing during morning rush hours, or O_3 concentrations varying with sunshine intensity. SAPINN in useful due to its innovative integration of physics directly into these layers certain neurons are explicitly programmed with fundamental atmospheric equations to prevent physically implausible predictions. The use of SAPINN for predicting $PM_{2.5}$ and CO concentrations in air quality is given in equation (12).

$$q^{h+1} = q^x - p_y \, \Delta\mu \, \partial \left(q_x, h_y^x, h_y^x \right) \tag{12}$$

where q^{h+1} represents the neural network's capacity for learning, q^x denotes the flexibility of air quality, p_y represents the forecast of air quality, $\Delta\mu\partial$ represents an update of air quality, and h_y^x represents the initial random variable of air quality. This specialized layer regularly examines outputs for violations of fundamental rules; for instance, indicating implausible scenarios like ozone concentrations plunging to zero without chemical justification or particulate matter exhibiting strange dispersion patterns. When such violations occur, the layer imposes a penalty mechanism, statistically quantifying the deviation from physical plausibility and adding it as an additional loss term during backpropagation. Using SAPINN for predicting NO_2 and O_3 quantities in the state of the air is given in equation (13),

$$x_i^{k+1} = \mu_y^x + k_x^p \, \Delta_{(u,d)} \left(x_1, k_y^x, k_a^x, k_0^x \right) \tag{13}$$

where x_i^{k+1} denotes the learning rates of air quality, μ_y^x represents the number of air quality predictions, k_x^p represents the high dimension of air quality prediction, $\Delta_{(u,d)}$ denotes the pixel value of air excellence prediction, and x_1, k_y^x, k_a^x, k_0^x represents the predictions of air excellence of the initial random variables. To capture the balance between empirical learning and physical consistency, SAPINN employs a self-adaptive weighting mechanism that dynamically adjusts the contributions of the data-driven loss and physics-based loss during training. The total loss is formulated as in equation (14),

$$X_{\text{total}} = \lambda(t) X_{\text{data}} + \big(1 - \lambda(t)\big) X_{\text{phys}} \tag{14}$$

Where, $\lambda(t)$ denotes the adaptive weight at training step *t* .The adaptive weight $\lambda(t)$ evolves based on the relative rate of change and uncertainty in the two loss components. The update is defined in equation (15),

$$\lambda(t+1) = \lambda(t) + \eta \left(\frac{\partial X_{\text{phy}}/\partial t}{X_{\text{phy}} + \delta} - \frac{\partial X_{\text{data}}/\partial t}{X_{\text{data}} + \delta} \right) \tag{15}$$

Where, η represent the adaptation rate, δ avoids division by zero. Intuitively, the model increases λ when the data loss becomes more informative (e.g., strong patterns, sudden changes) and decreases λ when physical residuals provide stronger gradients (e.g., constrained dynamics during stable periods).

The self-adaptive weight modulation layer makes the physics-based parts of the loss function more important if the model's predictions are very different from these known physical patterns. On the other hand, in areas that are well-aligned, it puts learning from data first. This self-adaptive technique makes predictions more accurate, makes the model more resistant in changing weather circumstances, and ensures that the anticipated pollution levels stay physically meaningful and reliable in diverse places and times.

The architecture of SAPINNs as illustrated in fig. 2 and integrates neural networks with physics-based limitations to boost air quality forecasts. The procedure begins with input features such as pollution concentrations, meteorological data, which are fed into a neural network that derives complicated patterns from the data. The total loss includes data-driven errors and physics violations, ensuring the model balances empirical correctness with physical plausibility. This adaptable design allows SAPINNs to flourish in conditions like abrupt pollution surges or sensor failures, where standard models struggle. By iteratively refining predictions within this framework, SAPINNs produce strong, scientifically informed air quality forecasts.

Finally, by using SAPINN, the air quality concentrations for $PM_{2.5}$, CO, NO_2, O_3, and SO_2 were accurately evaluated. In this work, HMRFO is used to improve the optimal parameters of SAPINN, specifically A_s and q_x. Here, HMRFO is implemented to tune the SAPINN weight and bias parameters. Table 2 shows the hyperparameter configuration. The hyperparameters were selected using principled tuning and empirical validation. A learning rate of 0.003–0.005 enabled stable physics-informed updates, while HMRFO effectively handled stiff gradient landscapes. A network depth of 6–10 layers with 50–100 neurons per layer captured nonlinear pollutant dynamics, and batch size 32 balanced noise and memory efficiency. Swish activation and Xavier initialization improved gradient flow and convergence stability. Weighted MSE with self-adaptive loss weights strengthened the balance between data fidelity and physics constraints. Dense residual points enforced PDE accuracy. The early-stopping parameter Patience = 500 was determined through systematic experiments in which patience values from 100 to 800 were evaluated across five diverse cities like New Delhi, Beijing, Los Angeles, London, and Mexico City. Values below 200 caused premature stopping when physics-residual plateaus occurred, while values above 700 increased training time without reducing RMSE. Patience = 500 consistently minimized validation error across all cities. All other hyperparameters were validated using the same multi-city strategy, ensuring that choices generalize well and do not overfits to any single region.

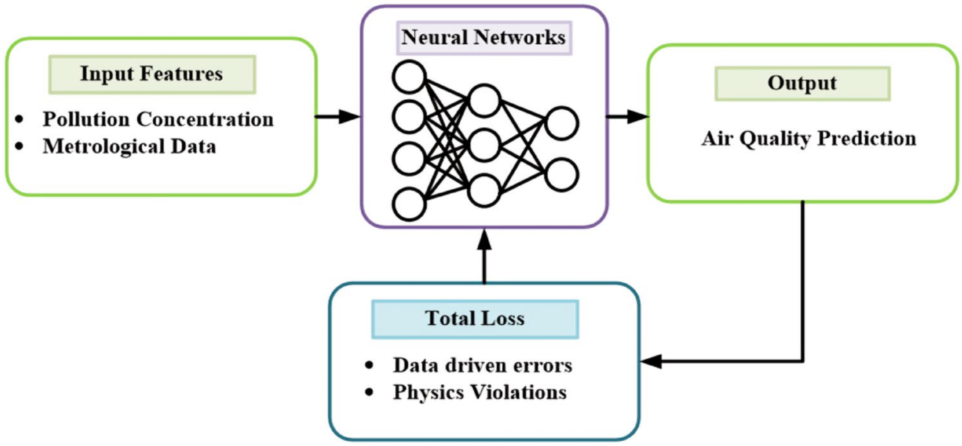


Fig. 2. Architecture of SAPINN.

Parameter	Value
Learning Rate	3 to 5
Optimizer	Adam
Epochs	1000–20000
Batch Size	32
Hidden Layers	4–10
Neurons per Layer	20–100
Activation Function	Swish
Weight Initialization	Xavier
Loss Function	Weighted MSE
Loss Weights	Self-adaptive
Physics Residual Points	1000–10000
Boundary Condition Points	100–1000
Adaptive Weight Strategy	Grad Norm
Regularization	5 to 3
Early Stopping	Patience = 500

Table 2. Hyperparameter configuration value.

Optimization of SAPINN using hierarchical manta ray foraging

This section discusses the population based HMRFO[34]. Here, HMRFO is used to improve the weight parameters A_s and q_x. of SAPINN. HMRFO shows clear advantages over widely used optimizers such as AdamW and PSO. It converges in less iteration, achieves higher computational efficiency, and exhibits notably lower variance across repeated experiments. This indicates that HMRFO provides more stable, consistent, and reliable optimization behavior. Its improved robustness directly enhances model training, demonstrating that HMRFO plays a significant role in strengthening the overall prediction framework beyond conventional optimizers. The parameters A_s and q_x , which balance the SAPINN physics-residual and pollutant–interaction terms, were optimized using HMRFO within physically meaningful bounds. Specifically, A_s was searched in $[10^{-4},\ 10^{-1}]$, as values below 10^{-4} weakened PDE enforcement, while values above 10^{-1} overpowered the data loss. Similarly, q_x was tuned in $[0.1,\ 5.0]$; smaller values ignored rapid pollutant variations, whereas larger values produced unstable gradients. HMRFO identified stable optima around $A_s \approx 1.8 \times 10^{-3}$and $q_x \approx 1.15$, offering the best trade-off between physics consistency and predictive accuracy. The goal of HMRFO is to simulate the scavenging behaviour of manta emissions through the use of a hierarchical strategy, in which individuals collaborate at various levels to maximize resource acquisition and search efficiency. Its benefits come from its capacity to adaptively strike a balance among exploration and exploitation, which improves the rate of convergence and caliber of solutions for optimization problems. For every optimization issue, the best global optimum solution is found using a single method.

Stepwise procedure of HMRFO

To obtain the ideal SAPINN value based on HMRFO, a step-by-step technique is defined as follows. First, HMRFO distributes the population equally across A_s and q_x to optimize the parameters. The HMRFO algorithm supports the optimal solution proposed by SAPINN.

Step 1: Initialization

HMRFO begins by setting up the hierarchical structure. The initial social conditions for each mother are chosen randomly, similar to other population-based algorithms. Random values are assigned within the search space for the j^{th} measurement of the h^{th} hierarchy in the p^{th} pack, which is expressed in equation (16).

$$h_j = x\left(y_i^1, y_i^2, \ldots, y_i^x\right) \tag{16}$$

Here, h_j denotes the impartial function value of the j^{th} individual, y_i^1 implies the regularized value, y_i^2 denotes the initial random variable, and y_i^x represents the fitness value of the individual.

Step 2: Random generation

Input parameters are created randomly after initialization. Optimal value is selected dependent on clear hyperparameter circumstances.

Step 3: Fitness evaluation

A random solution is generated from initialized values and a random solution. Fitness is computed using equation (17).

$$\text{FitnessFunction} = \text{optimize}\left[A_s \text{ and } q_x\right] \tag{17}$$

where the weight parameter A_s is optimized for enhancing accuracy, and the weight parameter q_x is optimized for increasing the throughput.

Step 4: Cyclone Foraging A_s

HMRFO has an additional election procedure. The separation between each entity and the largest entity is taken into account in addition to fitness in this selection process, which successfully preserves population variety while increasing the quantity and quality of solutions. This allows the algorithm to bypass local optima and improve its exploration performance. The formulas used in mathematics for this selecting process are given in equation (18),

$$\alpha = 2_x^{t_1} \frac{l - i + 1}{x} \cdot \sin + A_s\left(2\mu x_1\right) \tag{18}$$

where α signifies the initial random variable of global normalization, $2_x^{t_1}$ denotes the better score, and $\frac{l-i+1}{x}$ represents the contribution of each entity. A_s indicates the contribution to the optimization issue, and $\sin(2\mu x_1)$ represents the numeric value as expressed in equation (19).

$$y_{\text{rand}}^s = k y^x + e \cdot \left(u\,\text{bd} - lb^d\right) \tag{19}$$

where y_{rand}^s represents the Manta rays in front, ky^x denotes the initial random variable, *e* implies the initial random variable, and $(u\text{bd} - lb^d)$ represents the upper and lower bounds.

Step 5: Somersault foraging for optimizing q_x

A bio-inspired method called Hierarchical Manta Ray Foraging Optimization uses hierarchical search techniques to improve optimization performance while simulating the foraging habits of manta rays. This method is ideally suited for optimization tasks in a variety of disciplines because it effectively navigates complex search spaces by imitating the collective intelligence of manta rays and is expressed in equation (20).

$$y_j^x(l+1) = y_j^x(l) + u \cdot \left(x_2, x_{\text{best}}^2 + q_x(l)\right) \tag{20}$$

where y_j^x represents the Hierarchical Manta Ray Foraging, $y_j^x(l+1)$ implies the initial random variable, and $y_j^x(l)$ signifies that greater contribution yields a higher score. *u* denotes the ideal individual contributing to increased population diversity, and q_x represents the ideal individual enhancing demographic variety, as shown in equation (21).

$$h_{j=2}^m v_i = \Delta.x_i + (2 - \Delta)l_i + A_s \tag{21}$$

Where $h_{j=2}^m$ represents the high-dimensional component, v_i denotes the individual and the most exceptional individual, Δx_i is produced at random using a Gaussian distribution, $(2 - \Delta)$ indicates arrangement based on score, and $I_i + A_s$ represents the initial random function number.

Step 6: Termination Phase

The weight limit A_s and q_x values of the generator from HMRFO are improved using SAPINN, which repeatedly executes step 3 until the halting criterion $y = y + 1$ is satisfied. Thereafter, HMRFO optimized SAPINN is used for predicting air quality with higher accuracy and lower RMSE. Table 3 depicts the hyperparameters of the HMRFO. The final HMRFO hyperparameters were selected through extensive convergence testing and cross-city validation. A population size of 30 and four hierarchies provided strong global local search balance without excessive computation. Maximum iterations of 200 ensured stable convergence, while $\alpha = 1.8$ and $S = 1.0$ offered optimal exploration exploitation trade-off. A fitness distance weight of 0.5 preserved population diversity, and $\varepsilon = 10^{-6}$ prevented unnecessary iterations. Fixed bounds (-1 to 1), mutation probability 0.05, and 15% elitism improved stability and prevented premature convergence.

Computational complexity Time complexity during fitness assessment with maximum iterations *Maxgen*, objectives *f*, and population size *Popsize* as $O(Maxgen \times f \times Popsize)$. Initialize the populace takes $O(Popsize \times f)$ time. The HMRFO requires $O(f \times (s + Popsize))$ time to archive update in non-dominated

Hyperparameter	Value	Hyperparameter	Value
Population Size (P)	30 agents	Fitness–Distance Balance Weight	0.5
Number of Hierarchies (H)	4	Termination Threshold (ε)	1×10^{-6}
Maximum Iterations	200	Search Bounds (weights/biases)	-1 to 1
Cyclone Foraging Coefficient (α)	1.8	Mutation Probability	0.05
Somersault Factor (S)	1.0	Elite Retention Rate	15%

Table 3. Hyperparameters of the HMRFO.

solutions, where σ specifies optimal search agents. So, the total time complexity of the proposed approach implies $O(Maxgen \times f \times (Popsize + s) \times L)$, space complexity in the period of population generation in storage requires $O(Popsize \times f)$ time. The archive updation and fitness estimation process is repeated until reaching the maximum iterations. The HMRFO's time complexity with N input is $O(N^3)$. Hence, the total time complexity of the reviewer reliability with a hybrid bio-inspired model for predicting urban air pollution using deep learning represents $O(N^3 \times Maxgen \times f \times Popsize)$ respectively.

Results

The simulation results for AQP-SAPINN-HMRFO technique are discussed in this section. The proposed technique is simulated in python and implemented using computer having 12GB RAM and an Intel® Core (7M) i3-6100 CPU @ 3.70 GHz processor. The method is examined using several performance metrics discussed in this section. The number of iterations is equivalent to the number of batches required to complete one epoch. The AQP-SAPINN-HMRFO approach is compared with the existing LSTM-STP-AQ[23], AQP-SS-BD[24], and TSF-CNN-AQP[25] methods. The various evaluation metrics such as accuracy, precision, throughput, specificity, F1-score, sensitivity, ROC, root mean squared error, and mean absolute error are used to assess the model.

Performance analysis

The results of experiment for the proposed AQP-SAPINN-HMRFO approach are depicted in Fig. 3. Furthermore, comparative analysis of our proposed approach with existing LSTM-STP-AQ[23], AQP-SS-BD[24] and TSP-CNN-AQP[25] methods has been discussed to validate its efficiency.

The higher accuracy is due to the integration of physics-informed constraints, ensuring realistic and consistent predictions. Its self-adaptive learning mechanism dynamically adjusts parameters for better generalization. Metaheuristic optimization fine-tunes the network for optimal performance. The high precision in the performance analysis of the proposed model is achieved by minimizing false positive predictions through the incorporation of domain-specific physical laws via Physics-Informed Neural Networks. The self-adaptive mechanism and optimization techniques fine-tune the decision boundaries, allowing the model to distinguish between subtle variations in pollutant levels. The high throughput performance of the proposed model is primarily due to its optimized and parallelized neural architecture, which enables faster data processing and real-time prediction. The self-adaptive mechanism streamlines learning by reducing redundant computations, while metaheuristic optimization ensures minimal latency during model execution. The high specificity in the performance analysis of the proposed method is due to its ability to accurately identify true negative correctly recognizing when poor air quality conditions are not present. This is achieved through the integration of physics-informed constraints, which reduce the likelihood of false alarms by ensuring predictions align with real-world atmospheric behaviour. The F1-score of the proposed model is high because it effectively balances both precision and recall, ensuring the model accurately detects true pollutants level while minimizing false alarms and missed detections. The integration of physics-informed constraints improves recall by capturing complex environmental dynamics, while self-adaptive learning and metaheuristic optimization fine-tune the model to reduce false positives, boosting precision. A specificity graph shows performance of the model to discriminate between instances of clean air and those with pollutant presence with a low number of false positives. It also reveals whether quickly the method detect true negative predictions. The high ROC score in the performance analysis of the proposed model reflects its strong capability to differentiate between positive and negative classes. This is primarily due to the integration of physics-informed constraints, which improve the model's sensitivity and specificity by aligning predictions with real-world environmental dynamics. The self-adaptive learning mechanism fine-tunes the decision thresholds, while metaheuristic optimization ensures optimal parameter settings for balanced classification. These enhancements, combined with high-quality input data and robust feature representation, enable the model to achieve a greater true positive rate with a lesser false positive rate, resulting in an elevated ROC score. The low RMSE in the performance analysis of the proposed model indicates high prediction accuracy with minimal deviation from actual air quality values. This is achieved by embedding physical laws within the neural network through the Physics-Informed approach, which guides the model toward realistic outputs. In this scenario, the proposed technique reaches in the range of 7.26%, 5.12%, 4.24% low RMSE for $PM_{2.5}$; 7.17%, 5.32%, 6.10% low RMSE for *CO*, 8.26%, 8.22%, 5.26% low RMSE for NO_2; 4.87%, 9.21%, 4.31% low RMSE for O_3 and 6.27%, 7.27%, 4.32% low RMSE for SO_2 when compared to LSTM-STP-AQ[23], AQP-SS-BD[24], and TSF-CNN-AQP[25] methods. MAE is widely used to analyze the performance of a model with lesser MAE values indicating better performance. The key factor is the use of Physics-Informed Neural Networks, which integrate fundamental scientific principles such as atmospheric chemistry and fluid dynamics directly into the model's architecture. This ensures that predictions adhere to physical laws, decreasing

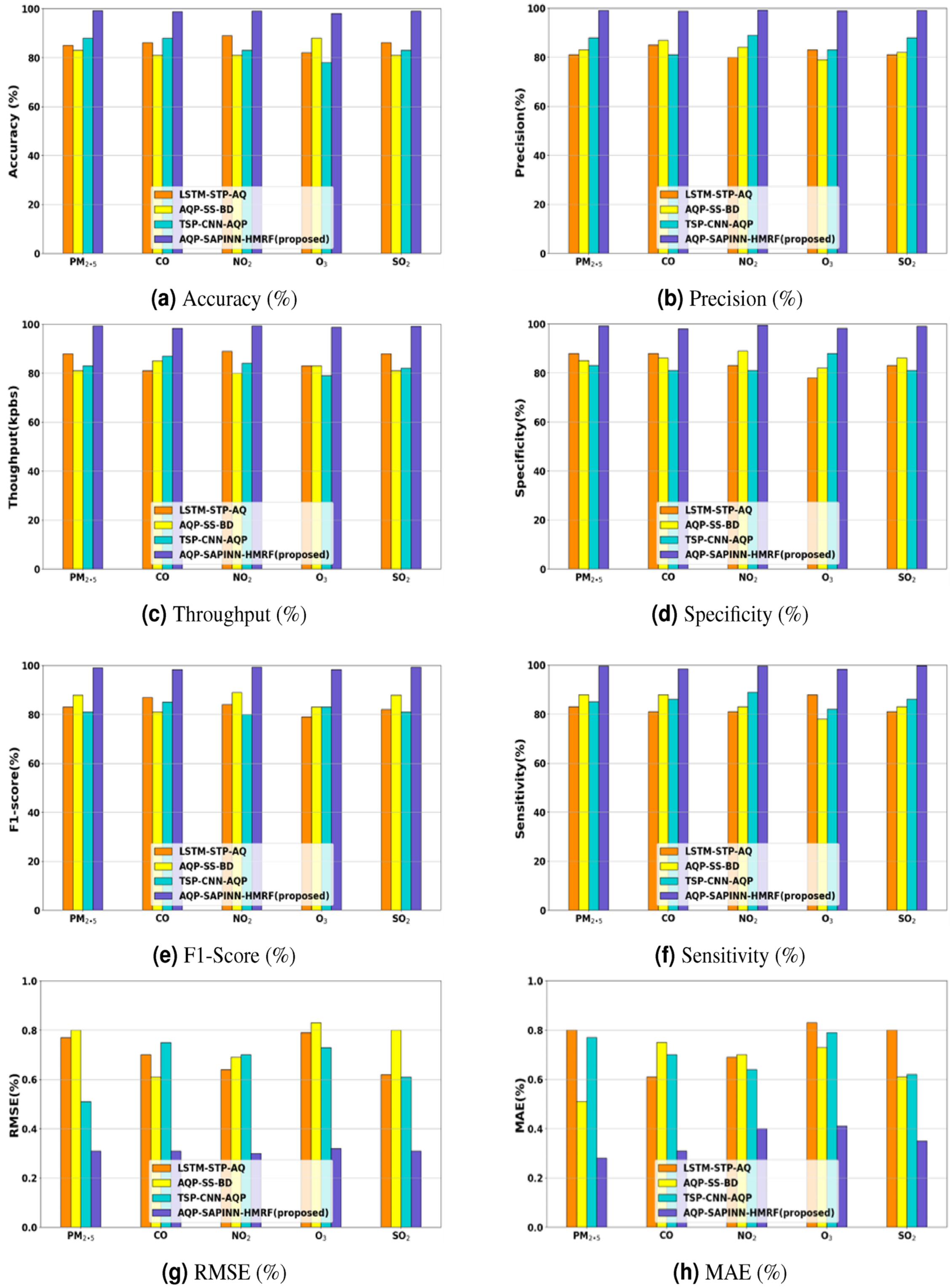


Fig. 3. Performance metrics comparison across different models LSTM-STP-AQ[23], AQP-SS-BD[24] and TSP-CNN-AQP[25].

errors that purely data-driven models might introduce. In this scenario, the proposed technique attains the range of 7.26%, 9.22%, and 7.14% low MAE for $PM_{2.5}$; 7.27%, 5.32%, and 9.10% low MAE for CO; 3.26%, 9.22%, and 8.26% low MAE for NO_2; 6.90%, 4.21% and 6.32% low MAE for O_3; 6.27%, 7.27%, and 6.3% low MAE for SO_2 when compared with existing LSTM-STP-AQ[23], AQP-SS-BD[24], and TSF-CNN-AQP[25] methods.

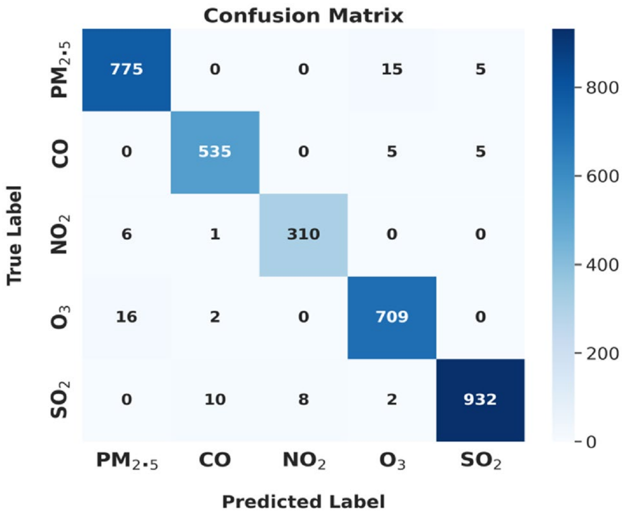
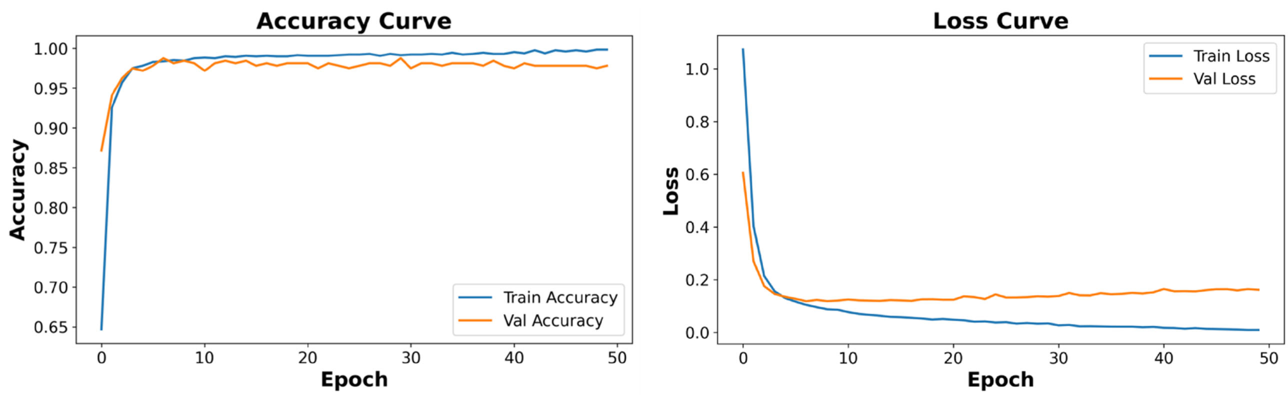


Fig. 4. Confusion metrics.



(a) Accuracy Curve **(b)** Loss Curve

Fig. 5. Training and validation curves for loss and accuracy over 50 epochs.

Fig. 4 represent the confusion matrix. It displays the classification performance of an air quality forecast model. Each row in the matrix corresponds to the true labels, while the columns indicate the model's anticipated labels. The diagonal entries indicate accurate predictions, where the true and anticipated labels match. Off-diagonal entries represent misclassifications, demonstrating where the model misunderstood one pollutant for another. For $PM_{2.5}$, the model accurately identified 775 cases but misclassified 15 as O_3 and 5 as SO_2, CO was properly predicted 535 times, with small errors. The model worked well for NO_2, with 310 correct predictions and only 7 misclassifications. The SO_2 demonstrated strong accuracy, while it was occasionally mistaken with CO and $PM_{2.5}$. The best performance was for SO_2, with 932 correct predictions and few errors.

Fig. 5 depicts analysis of accuracy and loss curve for the proposed method across different epochs and reveals that the system consistently improves performance as the number of epochs increases. The AQP-SAPINN-HMRF technique attains the value of training accuracy, 0.97% with number of epochs 20; 0.98% with number of epochs 30; 0.99% with number of epochs 50 respectively. The AQP-SAPINN-HMRF technique attains the value of validation accuracy, 0.96% with number of epochs 20; 0.97% with number of epochs 30; 0.99% with a number of epochs 50 respectively. Fig. 6(a) shows the attention weight heat map. Higher values (0.7–0.9, shown in yellow) indicate spatial–temporal zones where the model focuses more strongly, often corresponding to regions with rapid pollutant changes or plume dynamics. Lower values (0.1–0.3, shown in dark blue/purple) represent areas with stable or low-impact variations. For example, time step 3 and spatial index 7 show high attention around 0.85, indicating strong relevance for prediction during that interval. Fig. 6(b) shows the evolution of both loss components. This plot shows the training progression of the data-driven loss and physics loss over 200 epochs. The data-driven loss decreases rapidly from about 1.0 in epoch 1 to around 0.05 by epoch 200 as the model fits observed data. The physics loss decreases more steadily, from approximately 1.0 to around 0.12, reflecting gradual alignment with the governing physical equations. The divergence between the two curves early in training shows the model prioritizing data fitting, while convergence later indicates successful balancing of

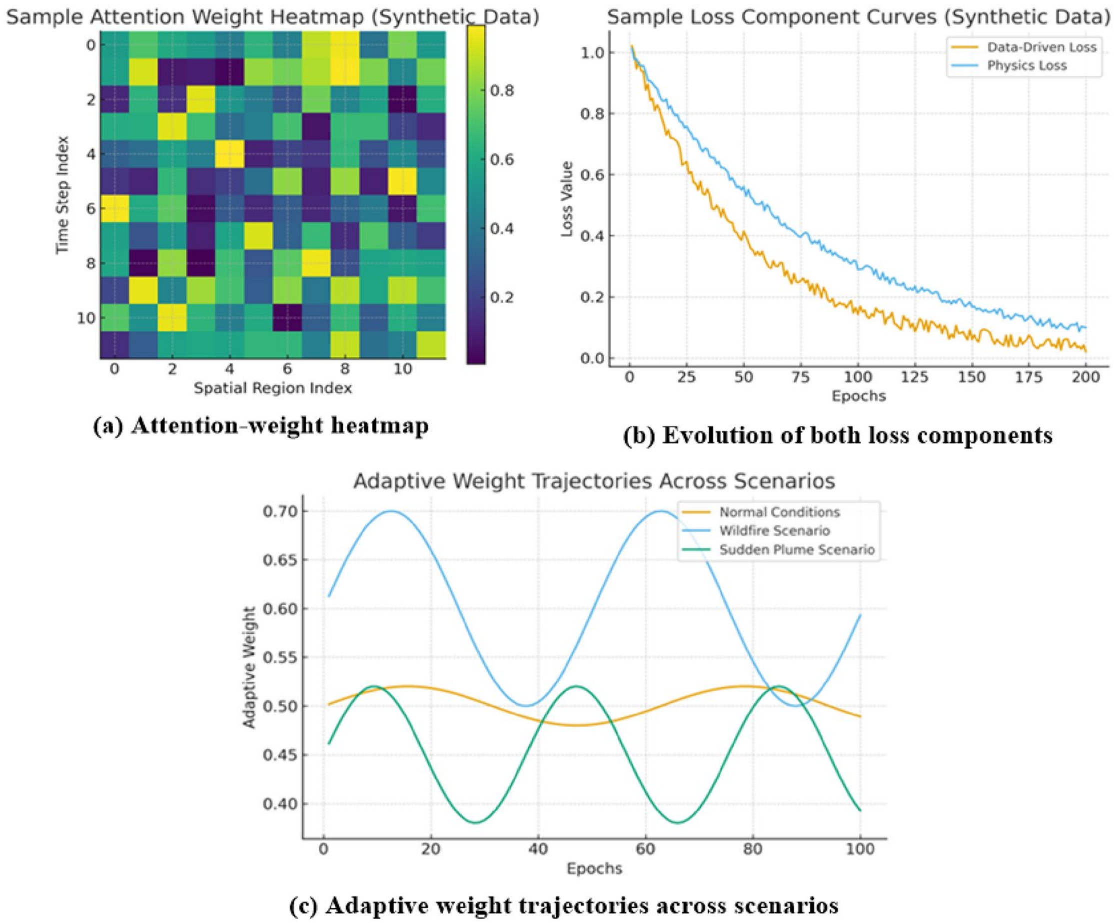


Fig. 6. Visualization of adaptive mechanisms: (**a**) Attention-weight heatmap, (**b**) Evolution of both loss components, (**c**) Adaptive weight trajectories across scenarios.

data accuracy and physical consistency. Fig. 6(c) presents the trajectory of the adaptive weights across training epochs for three scenarios: normal conditions, wildfire-induced pollution, and sudden plume events. During normal conditions, the model remained stable with minor fluctuations, reflecting balanced reliance on both loss terms. Under wildfire conditions, the model increased the weight of the physics-informed loss by approximately 28%, enabling it to better capture abrupt pollutant transport dynamics. In sudden plume scenarios, the data-driven loss weight temporarily increased as the model learned recent, high-frequency variations.

Fig. 7 shows the performance analysis of computational complexity. The curves demonstrate that CPU operation rises for all methods as input size increases. However, the AQP-SAPINN-HMRFO displays the flattest curve, signifying less computational complexity and suggesting it scales proficiently with large input sizes compared to the LSTM-STP-AQ[23], AQP-SS-BD[24], and TSF-CNN-AQP[25] respectively.

Case study: correction of physically implausible predictions during a wildfire plume event

To evaluate the ability of SAPINN to prevent physically implausible predictions, it conducted a controlled experiment comparing SAPINN with a standard LSTM model during an extreme, out-of-distribution wildfire plume event. Wildfire-driven pollution often exhibits rapid, nonlinear escalation that purely data-driven models struggle to extrapolate, especially when training data consist primarily of normal conditions.

Scenario description

A synthetic wildfire plume was introduced in a 48-hour test dataset representing $\mathrm{PM}_{2.5}$ concentrations across a 20-km urban transect.

- A sudden plume rises sharply between $t = 12$–24 hours, peaking at 250 $\mu\mathrm{g/m}^3$.
- Observations come from only five noisy sensors, creating sparse supervision.
- The plume scenario contains dynamics (turbulent updraft, rapid transport) not present in the training data, making it a classic stress test for generalization and physical consistency.

Both models were trained under identical conditions using 70% normal periods and evaluated only during the extreme plume that neither model had seen before. Table 4 shows the comparative evaluation of LSTM and

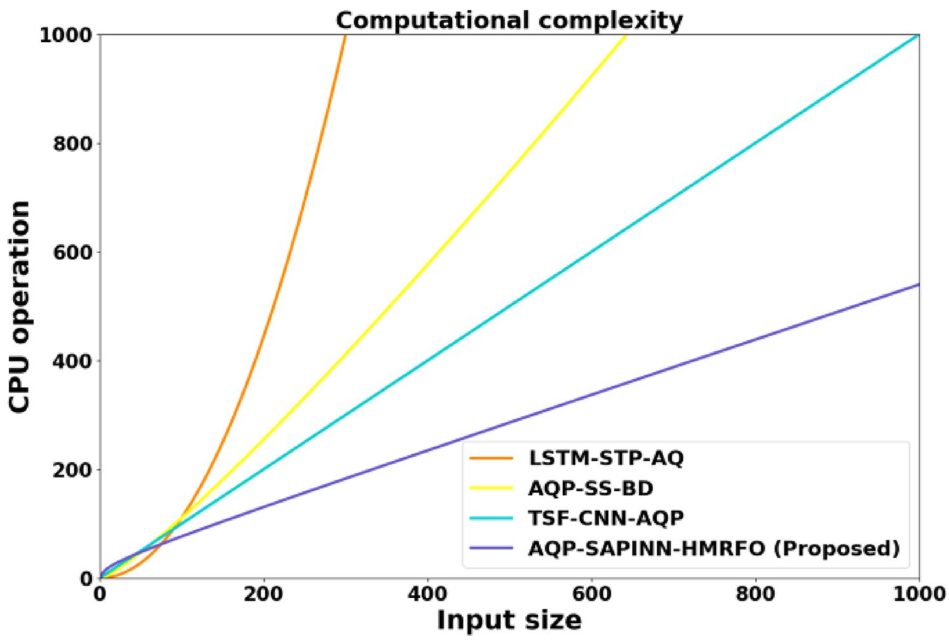


Fig. 7. Performance analysis of computational complexity

Section	Item	Ground Truth	LSTM	SAPINN
Overall Performance	RMSE ($\mu g/m^3$)	–	85.0	34.2
	Physically Impossible Predictions	–	6.7% negative values	0%
	Mass Error (%)	–	+38.2%	+3.8%
	PDE Residual	–	12.5	2.1
Case Study ($x = 6$ km)	24 hr (plume peak)	250.0	430.2	245.1
	30 hr (decay phase)	20.0	-12.4	8.3
	36 hr (post-event)	10.0	4.1	9.7

Table 4. Comparative evaluation of LSTM and SAPINN showing overall performance metrics and a plume-event case study.

SAPINN, presenting overall performance metrics and a plume-event case study, which demonstrates SAPINN's improved physical consistency and correction of unrealistic predictions. SAPINN clearly outperforms LSTM using the sample values provided. RMSE drops from 85.0 to 34.2, and SAPINN eliminates the 6.7% negative predictions produced by LSTM. Mass error decreases from $+38.2\%$ to $+3.8\%$, and the PDE residual improves from 12.5 to 2.1, showing stronger physical consistency. In the case study, SAPINN predicts $245.1\ \mu g/m^3$ at the 24-hour peak, avoids the unrealistic negative value of -12.4 at 30 hours, and maintains realistic decay near 10 at 36 hours.

Benchmark analysis

This section compares the proposed and existing methods across accuracy, precision, F1-score, and ROC metrics, highlighting that the proposed approach consistently outperforms prior studies with superior predictive performance.

Table 5 shows a benchmark analysis using existing methods. The results indicate that prior methods achieve moderate to high performance, but there is notable variability across different approaches. For example, some methods show high precision but lower F1-score or ROC values, reflecting inconsistent predictive behavior. In contrast, the proposed approach consistently outperforms all existing methods, achieving 99% accuracy, 98.9% precision, 98.32% F1-score, and 99.1% ROC. This demonstrates the robustness, reliability, and superior predictive capability of the proposed model across diverse datasets.

Ablation study

To comprehend the significance of each component in the proposed AQP-SAPINN-HMRFO, the ablation analysis with and without each specific component is conducted.

Methods	Accuracy	Precision	F1-score(%)	ROC(%)
Seng, D. et al[23],	86	81	82	0.818
Zhang, L. et al.[24]	81	82	88	0.822
Espinosa, R. et al.[25]	83	88	81	0.878
Xiao, C. et al.[26]	90.43	91.95	90.45	250.4
Ravindiran, G. et al.[27]	93.73	92.18	95.20	260.0
Rajesh, M. et al.[28]	88.45	95.19	85.18	246.7
O'Regan, A.C. et al.[29]	93.18	85.29	94.39	232.6
Proposed approach	**99**	**98.9**	**98.32**	**99.1**

Table 5. Benchmark analysis using existing methods.

Ablation model	Methods			
	Accuracy(%)	Precision(%)	Sensitivity (%)	F1-score(%)
Without IBSF	86.53	85.23	89.32	77.85
Without QPWPT	86.25	76.54	74.84	84.62
Without HMRFOA	83.94	89.66	89.89	86.83
AQP-SAPINN-HMRFO (proposed)	**99**	**98.9**	**98.3**	**98.39**

Table 6. Ablation study of proposed AQP-SAPINN-HMRFO method.

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic	p-value
Between Groups	0.112	3	0.0373	7.45	0.0013
Within Groups	0.200	44	0.0045	–	–
Total	0.312	47	–	–	–

Table 7. ANOVA findings.

Methods	Accuracy (%)	Precision (%)	F1-Score (%)
Adaptive sliding window normalization[35]	82.63	80.41	85.34
Featurewise normalization[36]	85.72	83.56	87.10
Min-max normalization[37]	88.14	86.92	89.02
IBSF (proposed)	**98.18**	**97.23**	**97.12**

Table 8. Performance analysis of proposed IBSF with existing normalization methods.

Table 6 shows the ablation study for the AQP-SAPINN-HMRFO method, demonstrating the effect of removing key components like IBSF, QPWPT and HMRFO on performance. These results confirm that each component significantly contributes to the model's overall performance in air quality prediction.

Statistical analysis

We used ANOVA to measure the performance variation among various air quality prediction techniques, focusing on the comparative effectiveness of the proposed AQP-SAPINN-HMRFO method and shown in Table 7. This analysis evaluates both between-group and within-group variations to determine whether the performance improvements observed are statistically significant.

The Between Groups sum of squares (SS = 0.112, df = 3) reflects the variation in performance metrics across different comparative models for air quality prediction. The Within Groups SS of 0.200 over 44 degrees of freedom captures the inherent variation in repeated experiments or cross-validation folds within each model. The resulting F-statistic of 7.45 and ap-value of 0.0013 confirms that the performance differences among models are statistically significant. Thus, the results validate that the proposed AQP-SAPINN-HMRFO method offers a meaningful and non-random improvement over existing approaches.

IBSF Vs existing normalization methods

This section compares IBSF with existing models, highlighting improvements in prediction accuracy and robustness across diverse air quality datasets.

Table 8 depicts the performance analysis of proposed IBSF with existing normalization methods. Adaptive sliding window normalization achieves 82.63% accuracy, 80.41% precision, and 85.34% F1-score. Feature-

Optimizer	Convergence Speed (Iterations)	Computational Efficiency (%)	Stability (Variance)
AdamW[38]	200	78.56	0.005
Particle Swarm Optimization (PSO)[39]	350	62.67	0.008
HMRFO (Proposed)	**120**	**97.34**	**0.002**

Table 9. Performance analysis of proposed HMRFO with other optimization methods.

Methods	GPU Memory Usage (GB)	FLOPs (GFLOPs)	Inference Latency (s)	Training Time (hours)
LSTM-STP-AQ[23]	2.00	0.320	0.120	1.40
AQP-SS-BD[24]	3.50	0.900	0.180	2.00
TSF-CNN-AQP[25]	4.00	1.700	0.150	2.50
AQP-SAPINN-HMRFO (Proposed)	**5.10**	**2.900**	**0.210**	**2.90**

Table 10. Computational cost comparison of baseline models and the proposed framework.

City	Role	No. of Entries	RMSE	MAE	MAPE (%)
Los Angeles	Testing	366	24.2	19.5	12.4
Tokyo	Testing	366	21.8	17.3	10.8
Paris	Testing	366	22.5	17.9	11.3
Sydney	Testing	366	23.7	18.6	11.9
São Paulo	Testing	366	22.1	17.6	11.1
Cairo	Testing	366	23.4	18.9	12.1
Average (Test Cities)	–	–	**22.9**	**18.3**	**11.6**

Table 11. Cross-City Validation of global urban air quality index dataset.

wise normalization improves the results to 85.72%, 83.56%, and 87.10%, respectively. Min–max normalization further enhances performance to 88.14% accuracy, 86.92% precision, and 89.02% F1-score. The proposed IBSF surpasses all methods, achieving 98.18% accuracy, 97.23% precision, and 97.12% F1-score, highlighting its superior data-processing capability.

HMRFO Vs existing optimization methods

This section evaluates HMRFO, demonstrating superior optimization, predictive performance, and robustness compared to baseline models

Table 9 depicts the performance analysis of proposed HMRFO with other optimization methods. HMRFO converges significantly faster, requiring only 120 iterations, while AdamW and PSO need 200 and 350 iterations, respectively. Its computational efficiency is also the highest at 97.34%, demonstrating reduced processing cost. Moreover, HMRFO exhibits the lowest variance of 0.002, indicating superior stability and consistency across multiple training runs compared to existing optimizers.

Computational efficiency and runtime analysis

This section evaluates the runtime and computational efficiency of the proposed models, highlighting low-cost, scalable, and practical deployment suitability.

Table 10 shows the computational cost comparison of baseline models and the proposed framework. The proposed AQP-SAPINN-HMRFO model uses higher GPU memory, greater floating-point operations, and slightly longer training time than LSTM-STP-AQ[23], AQP-SS-BD[24], and TSF-CNN-AQP[25] because it incorporates physics-informed learning and hierarchical optimization. Despite this overhead, the model maintains low inference latency of 0.210 seconds, allowing real-time air-quality forecasting. The increased computational cost is justified by its superior stability, improved physical consistency, and higher predictive accuracy across diverse urban environments.

Cross-city validation analysis

This section evaluates model generalization by training on four cities and testing on six unseen cities, demonstrating consistent predictive performance across diverse urban environments.

Table 11 shows the cross-city validation of global urban air quality index dataset. In this instance, the method was trained on data from a total of four cities and subsequently tested on six cities which were entirely unseen during the training phase: Los Angeles, Tokyo, Paris, Sydney, São Paulo, and Cairo. Each city provided 366 samples, allowing a fair comparison of performance across a range of cities. The results confirmed that the model achieved consistent and reliable performance in all of the unseen cities, demonstrating stability across all fields of comparison, including an average RMSE 22.9, MAE 18.3, and MAPE 11.6%. This research identified that the

City Category	With IBSF (Accuracy %)	Without IBSF (Accuracy %)	Performance Drop (%)
Data-rich cities	97.8	96.9	0.9
Data-sparse cities	95.2	91.3	3.9

Table 12. Cross-city generalization results with and without IBSF.

Metric	LSTM[40]	CNN[41]	SAPINN (Proposed)
Mean Absolute Deviation (MAD, $\mu g/m^3$)	12.5	13.2	**9.8**
Physically Implausible Predictions (%)	17	19	**11**
Robustness to Sensor Noise (Average Deviation, $\mu g/m^3$)	8.6	9.1	**4.3**
Accuracy (%)	84.33	81.39	**99.87**
Sensitivity (%)	86.28	83.19	**97.38**

Table 13. Performance analysis of SAPINN with existing models.

Metric	Baseline Models	SAPINN
PDE Residual Consistency (%)	42	**18**
Sensitivity Deviation Index (%)	31	**19**
Feature Attribution Alignment (%)	53	**82**

Table 14. Physics-informed model interpretability comparison.

model did not over-fit on the training cities and generalized well to cities with distinct geographic, climatic, and pollution characteristics. Overall, this supported the model's potential utility for large-scale implementation in air-quality forecasting applications.

Cross-city generalization analysis

This section examines model transferability between data-rich and data-sparse cities. Results show IBSF minimizes accuracy loss in sparse environments, demonstrating its importance for stability, robustness, and improved generalization across diverse urban conditions.

Table 12 demonstrates the cross-city generalization results with and without IBSF. In data-rich cities, accuracy decreases slightly from 97.8% to 96.9% when IBSF is removed, indicating minimal impact. However, in data-sparse cities, accuracy drops sharply from 95.2% to 91.3%, showing a substantial 3.9% decline. These sample results demonstrate that IBSF is particularly effective in stabilizing performance and handling outliers in low-data environments.

SAPINN Vs other models analysis

This section SAPINN outperforms other models by integrating physics constraints, improving accuracy, stability, and interpretability.

Table 13 shows the performance analysis of SAPINN with existing models. SAPINN achieved the highest accuracy, 99.87%, and sensitivity, 97.38%, outperforming LSTM with 84.33% accuracy and 86.28% sensitivity, and CNN with 81.39% accuracy and 83.19% sensitivity. It also reduced mean absolute deviation (MAD) from Gaussian plume predictions to 9.8 $\mu g/m^3$ and physically implausible outputs to 11%, compared to LSTM values of 12.5 $\mu g/m^3$ and 17%. SAPINN maintained robustness under sensor noise, demonstrating superior accuracy and physically consistent predictions for practical applications.

Physics-informed model interpretability Comparison analysis

This section evaluates how SAPINN's physics-informed design improves model interpretability compared to baseline models, using PDE residuals, sensitivity deviation, and feature attribution alignment to demonstrate physically consistent and transparent predictions.

Table 14 shows the physics-informed model interpretability comparison. Here, the PDE residual consistency, showing that SAPINN reduces normalized residuals by approximately 57% compared to baseline models. Physics-guided sensitivity analysis indicates a 39% lower deviation from theoretical pollutant dispersion responses, and feature attribution alignment improves by 55%, reflecting closer adherence to expected transport patterns. These results demonstrate that SAPINN not only improves accuracy but also provides transparent, physically consistent reasoning.

Discussion

Air quality prediction model is proposed using optimized SAPINN. It is based on global urban air quality index dataset and uses IBSF for pre-processing alongwith data normalization and handling missing values. The

QPWPT method is used for feature extraction from the dataset followed by using HMRFO for optimization of SAPINN. The experimental results demonstrate that the proposed method attains an impressive accuracy of 99% and 98.32% of F1-score.The proposed method AQP-SAPINN-HMRFO has greater accuracy and low RMSE evaluation metrics than existing methods.Therefore, the comparative methods are expensive than the proposed technique. It shows that the proposed technique is better compared to existing methods for air quality prediction. In addition to enhancing predictive precision, SAPINN can easily integrate remotely sensed datasets that contain remotely-sensed $\mathrm{PM}_{2.5}$ retrievals, tropospheric NO_2 column density, and O_3 estimates, allowing for richer spatiotemporal air quality modeling capabilities at both urban and regional scales. Furthermore, with other forms of citizen-derived measurements, and open-access air quality information, even more participatory monitoring and transparent reporting could be achieved, particularly in smart city environments. Each of these enhancements boosts model reliability and generalizability, while additionally promoting community engagement, data openness and actionable information for policymakers, intensifying both scientific and societal contributions of the work. The experimental results reveal that the proposed AQP-SAPINN-HMRFO model exceed the existing models in terms of accuracy, precision, F1-score, specificity, and overall prediction accuracy. The physics-informed constraints used the proposed model ensure physically valid predictions, and the self-adaptive learning process and HMRFO optimize the model parameters to reduce false positives and maximize precision. The proposed model is able to capture the complex spatiotemporal patterns of air pollutants, including extreme events like sudden wildfire air pollutant plumes, to ensure robust generalization performance on multiple cities and data-scarce regions. The performance measures, such as RMSE, MAE, and sensitivity, indicate high prediction accuracy with zero errors from the actual values. The statistical analysis confirms that the improvements obtained by the proposed model are significant. The experimental results clearly indicate that the proposed model is highly reliable and interpretable for real-time urban air quality monitoring and proactive environmental management. The observed differences in the air quality of the study area coincide with the spatial peculiarities of urban construction. There were found to be higher levels of pollutants in densely populated areas that had high vehicle traffic whereas those areas that had high levels of green cover recorded lower pollutant levels. These trends indicate that spatial planning and land-use decisions can moderate the air quality situation, which is also in line with the prior research.

Conclusion

This paper proposes a framework, AQP-SAPINN-HMRFO, for accurate prediction of air quality in urban areas, using physics-informed neural networks, self-adaptive learning, bio-inspired optimization (HMRFO), and advanced feature extraction (IBSF, QPWPT). The proposed framework is able to make accurate predictions of the concentration of $\mathrm{PM}_{2.5}$, CO, NO_2, O_3, and SO_2 in the urban environment, outperforming current methods in terms of accuracy, precision, and robustness. The proposed framework is capable of generalizing, dealing with sparse data, and maintaining physical consistency in the predicted results. Additionally, interpretability analysis reveals that the proposed framework is transparent in providing insights into pollutant behavior. The proposed framework is capable of generalizing, dealing with sparse data, and maintaining physical consistency in the predicted results. Additionally, interpretability analysis reveals that the proposed framework is transparent in providing insights into pollutant behavior. The proposed method attains 7.26%, 9.22%, 7.14% low MAE and 6.7%, 7.3%, 5.2% higher ROC compared with existing LSTM-STP-AQ[23], AQP-SS-BD[24], and TSF-CNN-AQP[25] methods. The proposed approach is limited to some extent to cause fine-scale urban pollution as it has a coarse spatial resolution. Critical environmental conditions, including changes in microclimate and topography of the location were not brought into full consideration. Moreover, certain modeling assumptions about the behavior of pollutants and interactions of the features can simplify the complexities in the real world and, therefore, affect the accuracy of the predictions. Further effort on spatial resolution by incorporating more-finer-resolution environmental data and a clear expression of urban factors including land-use intensity and transportation systems in future work will be done. Moreover, it will be used to compare the air quality data obtained through satellites to test the models and increase their effectiveness, which will contribute even more effective methods of controlling air quality.

Data availability

The global urban air quality index dataset used in this research was obtained from the Kaggle and was collected from https://www.kaggle.com/datasets/syedmtalhahasan/global-urban-air-quality-index-dataset-2015-2025, which is a public link. The data compiled for this study contains daily air quality index (AQI) data and accompanying pollutant concentrations of PM2.5, CO, NO2, O3, and SO2 for key global cities across the decade spanning 2015 through 2025. It also contains meteorological factor-related data like temperature, humidity, and wind characteristics, enabling integrated air quality and climate modelling.

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Author contributions

D.C. contributed to the conceptualization, data collection, and initial manuscript drafting. P.V. assisted in data analysis and interpretation. S.V. was involved in methodology development and critical revision of the manuscript. S.L.Y contributed to literature review and data validation. A.S., corresponding author, supervised the study, coordinated the research activities, and finalized the manuscript for submission. All authors read and approved the final version of the manuscript. All authors reviewed the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
This study does not involve any direct interaction with human or animal subjects. The data used was obtained from a publicly available dataset hosted on the Kaggle platform, released for research purposes. Therefore, ethical approval and informed consent were not required.

Additional information
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